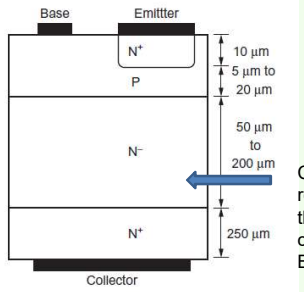
Lecture 5

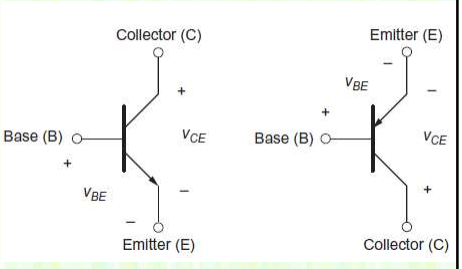
1

Power Electronic Devices- Power Bipolar Transistor

Bipolar junction transistors (BJT) are used to be the traditional component for power electronic applications. However, insulated gate bipolar transistor (IGBT) and metal-oxide-semiconductor field-effect transistor (MOSFET) technology have progressed so that they are viable replacements for the BJT. However, BJTs still have performance areas in which they may be preferably used, especially when the cost is the main concern.



Collector drift region. It increases the blocking capacity of Power BJT



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Power Bipolar Transistor



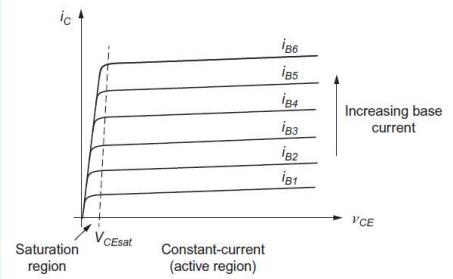



Basics of Power BJT

- It has large current handling and Power handling capacity compared to BJT .
- It offers high voltage resistance in OFF state then BJT .
- It offers high current handling in ON state then BJT .
- It has vertically oriented structure.
- High gain is maintained by enhancing doping level of emitter several times that of base .
- Due to more emitter doping, current gain(β) will decrease .
- It handles more current, so it has large metallic body to dissipate power as heat.

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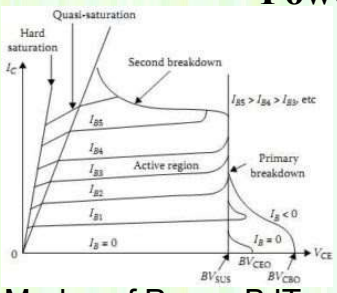
Power Bipolar Transistor



Switching power electronic system requires transistors to operate either in the saturation region where V_{CE} is small or in the cutoff region where the current is zero and the voltage is uphold by the device. In both situations, the dissipated power is low. The crossing of the active region must be as fast as possible to minimize the losses and the consequent temperature increase. The saturation region is characterized by the forward-biased of base-collector junction, and it occurs when V_{CE} <0:7V.

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Power Bipolar Transistor



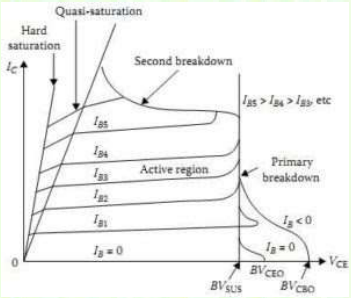
Vertical-structure power transistors have an additional region of operation called ***quasi saturation***, indicated in the characteristic curve of Figure.

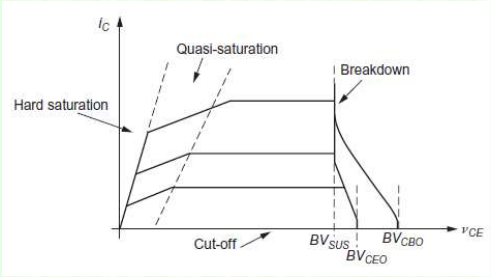
Modes of Power BJT

- Cut OFF mode [Base Emitter and Collector Base junction in Reverse Bias]
- Active mode [Base Emitter junction in Forward bias and Collector Base Junction in Reverse Bias]
- Quasi Saturation [Base Emitter and Collector Base junction In Forward Bias]
- Hard Saturation [Base Emitter and Collector Base junction in Forward Bias]

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Power Bipolar Transistor

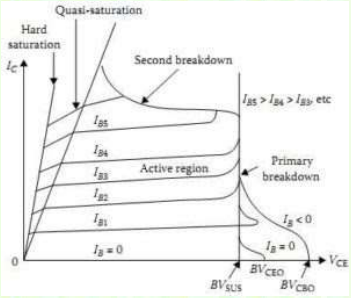


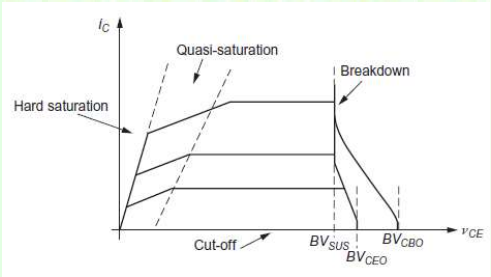


Concerning the voltage, there is a region called **primary breakdown** due to conventional avalanche of the collector-base junction and the attendant large flow of current. The BV_{SUS} is the limit for primary breakdown; it is the maximum collector emitter voltage that can be sustained across the transistor when it is carrying collector current. The BV_{SUS} is lower than BV_{CEO} or BV_{CBO} that measure the transistor's voltage standoff capability when the base current is zero or negative.

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Power Bipolar Transistor

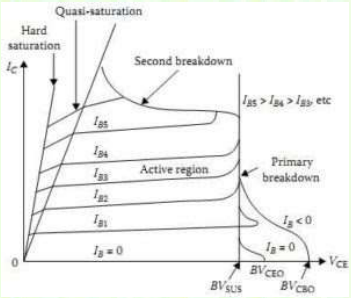


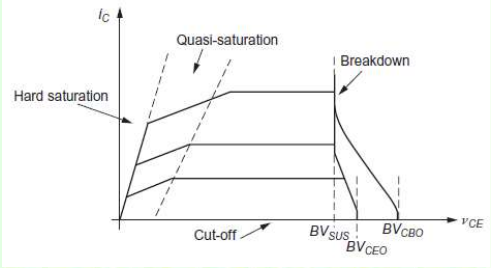


Power BJT has another potential failure mode called second breakdown, which shows as a precipitous drop in the collector-emitter voltage at large currents. Because the power dissipation is not uniformly spread over the device, but it is rather concentrated on regions, a local gradient of temperature can rise very quickly. Such thermal runaway brings hot spots that can eventually melt and recrystallize the silicon resulting in the device destruction.

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Power Bipolar Transistor





The key to avoid second breakdown is to
(1) keep power dissipation under control,
(2) use a controlled rate of change of base current during turnoff,
(3) Use of protective snubbers circuitry, and
(4) position the switching trajectory within the SOA boundaries

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Power BJT - Safe Operation Area

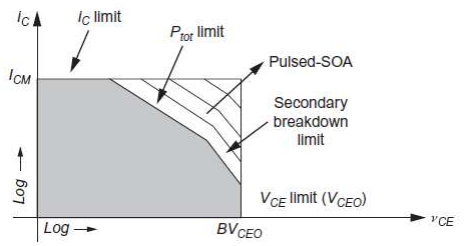


FIG. 2.24 Forward-bias safe operating area (FBSOA).

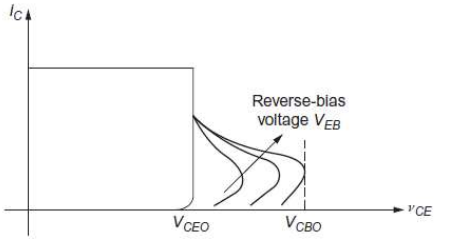


FIG. 2.25 Reverse-bias safe operating area (RBSOA).

In order to describe the maximum values of current and voltage the transistor should hold, two diagrams are used: the forward bias safe operating area (FBSOA), known also simply as SOA, given in Figure 2.24, and the reverse-bias safe operating area (RBSOA) shown in Figure 2.25

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Power BJT - Safe Operation Area

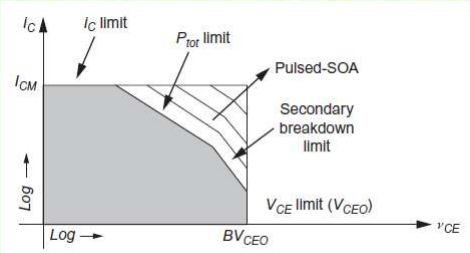


FIG. 2.24 Forward-bias safe operating area (FBSOA).

In the FBSOA, the current I_{CM} is the maximum current of the device; there is a boundary defining the maximum thermal dissipation and a margin defining the second breakdown limitation. Those regions are expanded for switching mode operation. Inductive load generates higher peak energy at turnoff than its resistive counterpart. It is then possible to have a secondary breakdown failure if RBSOA is exceeded.

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Power BJT - Safe Operation Area

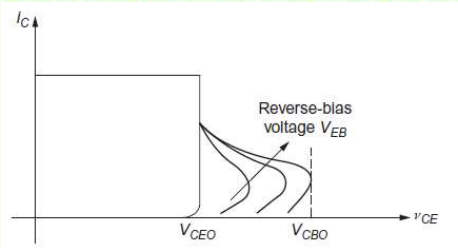


FIG. 2.25 Reverse-bias safe operating area (RBSOA).

A reverse base current helps the cutoff characteristics expanding RBSOA. The RBSOA curve shows that for voltages below V_{CEO}, the safe area is independent of reverse-bias voltage V_{EB} and is only limited by the device collector current, whereas above V_{CEO}, the collector current must be under control depending upon the applied reverse-bias voltage; in addition, temperature effects derate the SOA.

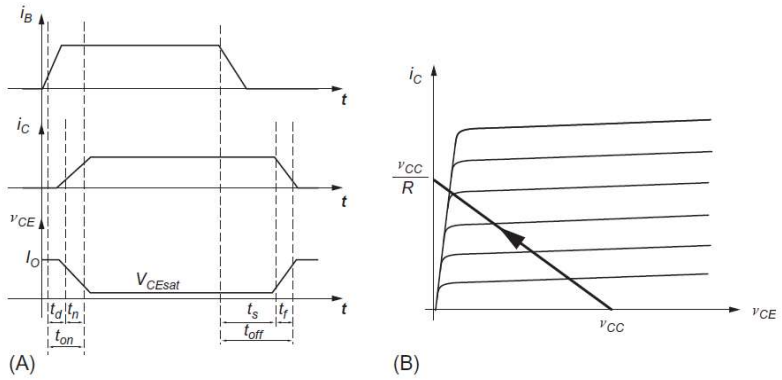
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Power BJT - Safe Operation Area

Ability for the transistor to switch high currents reliably is thus determined by its peak power handling capabilities. This ability is dependent upon the transistor's current and thermal density throughout the active region. In order to optimize the SOA capability, the current density and thermal density must be low. In general, it is a hot spot occurring at the weakest area of the transistor that will cause a device to fail due to second breakdown phenomenon. Although a wide base width will limit the current density across the base region, good heat sinking directly under the collector will enable the transistor to withstand high peak power. When the power and heat are spread over a large silicon area, these destructive tendencies are held to a minimum, and the transistor will have the highest SOA capability.

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Power BJT - Switching Characteristics



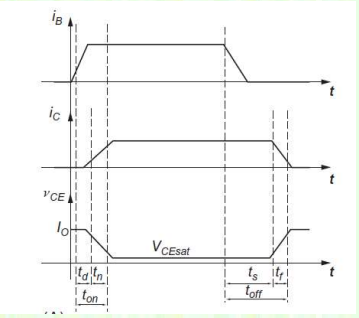
(A) Switching characteristics of a BJT under resistive load and (B) the trajectory over the static characteristic curves under resistive load.

t_f is the current falling time, that is, when the transistor is blocking such time corresponds to crossing from the saturation to the cutoff state.

Index "r" refers to the rising time (from 10 to 90% of maximum value);

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Power BJT - Switching Characteristics

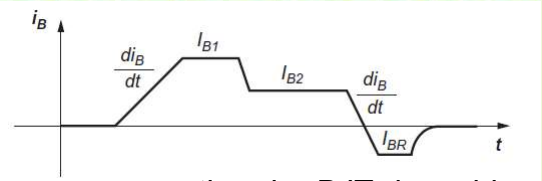


In order to reduce t_f, the base current for blocking must be negative, and the device must be kept in quasi saturation to minimize the stored charges. The delay time is denoted by t_d, corresponding to the time to discharge the capacitance of base-emitter junction, which can be reduced with a larger current base with high slope.

Storage time (t_s) is a very important parameter for BJT transistor. It is the required time to neutralize the minority carriers stored in the collector and base. Storage time and switching losses are key points to deal extensively with bipolar power transistors

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Power BJT - Switching Characteristics



Most advantageous operation in BJT is achieved when fast transitions are optimized. Such requirement minimizes switching losses. Therefore, a good bipolar drive circuit highly influences the transistor performance. A base drive circuit should provide a high forward base drive current (I_{B1}) as indicated in Figure to ensure the power semiconductor turn-on quickly. Base drive current should keep the BJT close to the saturation to minimize forward conduction losses, but a level I_{B2} would maintain the transistor in a quasi-saturation condition, avoiding excess of charges in the base. Controllable slope and reverse current I_{BR} sweeps out stored charges in the transistor base, speeding up the device turnoff

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Power Bipolar Transistor

Applications of Power BJT

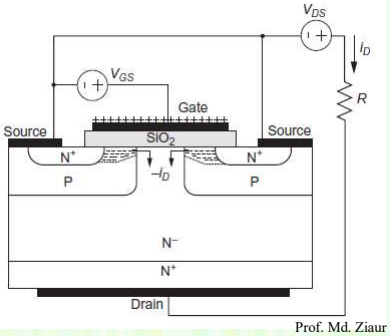
- SMPS
- Power Amplifier
- DC to AC inverters
- Relay
- Power control circuit

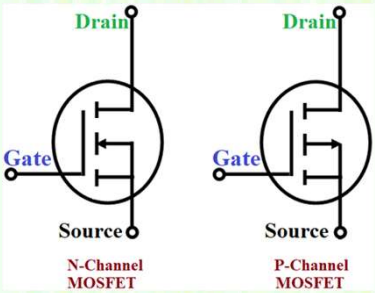
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Next Class

Power Electronic Devices- Power MOSFET

Power MOSFET is a type of MOSFET which is specially meant to handle high levels of power. These exhibit high switching speed and can work much better in comparison with other normal MOSFETs in the case of low voltage levels. However its operating principle is similar to that of any other general MOSFET. Power MOSFETs which are most widely used are n-channel Enhancement-mode or p-channel Enhancement-mode or n-channel Depletion-mode in nature.





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